

SAFETY DATA SHEET

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

Product Identifier

Perihalan Produk:

Chocolate Agar

Product Description:

Chocolate Agar

Cat No. :

R01293, R01300, R01301, R01302, R01302A, R01303, R02055, R060480, R060482, R060488, R08240

Relevant identified uses of the substance or mixture and uses advised against

Recommended Use

Laboratory chemicals.

Uses advised against

No Information available

Details of the supplier of the safety data sheet

Company

Thermo Scientific Microbiology Sdn Bhd
No.6, Jalan TTC 6, Taman Teknologi Cheng,
Cheng, 75250 Melaka, Malaysia
+606 334 0975 .

Supplier

Remel
12076 Santa Fe Drive Lenexa,
KS 66215 United States
Telephone: 1-800-255-6730
Fax:1-800-621-8251

E-mail address

mbd-sds@thermofisher.com

Emergency Telephone Number

(603) 5122 8888
CHEMTREC Malaysia **1-800-815-308** (Malay)
CHEMTREC Malaysia (Kuala Lumpur) **+(60)-327884561** (Malay)

SECTION 2: HAZARDS IDENTIFICATION

Classification of the substance or mixture

Label Elements

Hazard Statements

Precautionary Statements

Other Hazards

This product does not contain any known or suspected endocrine disruptors

SAFETY DATA SHEET

Chocolate Agar

Revision Date 29-Mar-2023

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

Component	CAS No	Weight %
Hydrogen chloride	7647-01-0	Trace
Octamethylcyclotetrasiloxane	556-67-2	Trace
Hemoglobins	9008-02-0	0.93
Glucose	50-99-7	0.07
Vitamin B12	68-19-9	Trace
L-Glutamine	56-85-9	Trace
Adenine (6-Aminopurine)	73-24-5	Trace
6H-Purin-6-one, 2-amino-1,7-dihydro-, monohydrochloride	635-39-2	Trace
p-Aminobenzoic acid	150-13-0	Trace
Cysteine hydrochloride, L-(+)-, monohydrate	7048-04-6	Trace
Thiazolium, 3-[(4-amino-2-methyl-5-pyrimidinyl)methyl]-4-methyl-5-(4,6,6-trihydroxy-3,5-di oxa-4,6-diphosphahex-1-yl)-, chloride, P,P'-dioxide	154-87-0	Trace
Iron(III) nitrate nonahydrate	7782-61-8	Trace
Thiazolium, 3-[(4-amino-2-methyl-5-pyrimidinyl)methyl]-5-(2-hydroxyethyl)-4-methyl- chloride, monohydrochloride	67-03-8	Trace
Adenosine 5'-(trihydrogen diphosphate), P'.fwdarw.5'-ester with 3-(aminocarbonyl)-1-.beta.-D-ribofuranosylpyridinium, inner salt	53-84-9	Trace
Peptones	73049-73-7	0.57
Sodium chloride	7647-14-5	0.24
Dipotassium phosphate	7758-11-4	0.34
Starch	9005-25-8	0.1
Propanoic acid, 2-oxo-, sodium salt	113-24-6	0.1
Yeast, ext.	8013-01-2	0.43
Caseins, hydrolyzates	65072-00-6	0.29
Gelatins, hydrolyzates	68410-45-7	0.29
Agar	9002-18-0	1.1
Water	7732-18-5	95.49

SECTION 4: FIRST AID MEASURES

Description of first aid measures

Eye Contact Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

Skin Contact Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur.

Ingestion Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.

Inhalation Remove to fresh air. Get medical attention immediately if symptoms occur.

Self-Protection of the First Aider No special precautions required.

Most important symptoms and effects, both acute and delayed

None reasonably foreseeable. . Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation.

SAFETY DATA SHEET

Chocolate Agar

Revision Date 29-Mar-2023

Indication of any immediate medical attention and special treatment needed

Notes to Physician Treat symptomatically.

SECTION 5: FIREFIGHTING MEASURES

Extinguishing media

Suitable Extinguishing Media

Water spray, carbon dioxide (CO₂), dry chemical, alcohol-resistant foam.

Extinguishing media which must not be used for safety reasons

No information available.

Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors.

Hazardous Combustion Products

None under normal use conditions.

Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

SECTION 6: ACCIDENTAL RELEASE MEASURES

Personal Precautions, Protective Equipment and Emergency Procedures

Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation.

Environmental precautions

Should not be released into the environment. See Section 12 for additional Ecological Information.

Methods and Material for Containment and Cleaning Up

Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

SECTION 7: HANDLING AND STORAGE

Precautions for Safe Handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid contact with skin, eyes or clothing. Avoid ingestion and inhalation. Avoid dust formation.

Conditions for Safe Storage, Including any Incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place.

Specific End Uses

Use in laboratories.

SAFETY DATA SHEET

Chocolate Agar

Revision Date 29-Mar-2023

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

Control Parameters

Component	Malaysia	ACGIH TLV	OSHA PEL
Hydrogen chloride		Ceiling: 2 ppm	Ceiling: 5 ppm Ceiling: 7 mg/m ³ (Vacated) Ceiling: 5 ppm (Vacated) Ceiling: 7 mg/m ³
Vitamin B12			(Vacated) TWA: 5 mg/m ³
Iron(III) nitrate nonahydrate		TWA: 1 mg/m ³	(Vacated) TWA: 1 mg/m ³
Starch		TWA: 10 mg/m ³	(Vacated) TWA: 15 mg/m ³ (Vacated) TWA: 5 mg/m ³ TWA: 15 mg/m ³ TWA: 5 mg/m ³

Component	European Union	The United Kingdom	Germany
Hydrogen chloride	TWA: 5 ppm (8h) TWA: 8 mg/m ³ (8h) STEL: 10 ppm (15min) STEL: 15 mg/m ³ (15min)	STEL: 5 ppm 15 min STEL: 8 mg/m ³ 15 min TWA: 1 ppm 8 hr TWA: 2 mg/m ³ 8 hr	TWA: 2 ppm (8 Stunden). AGW - exposure factor 2 TWA: 3 mg/m ³ (8 Stunden). AGW - exposure factor 2 TWA: 2 ppm (8 Stunden). MAK TWA: 3.0 mg/m ³ (8 Stunden). MAK Höhepunkt: 4 ppm Höhepunkt: 6 mg/m ³
Vitamin B12		STEL: 0.3 mg/m ³ 15 min TWA: 0.1 mg/m ³ 8 hr Resp. Sens. STEL: 15 mg/m ³ 15 min TWA: 5 mg/m ³ 8 hr Skin	TWA: 2 mg/m ³ (8 Stunden). MAK Höhepunkt: 2 mg/m ³ Haut
Iron(III) nitrate nonahydrate		STEL: 2 mg/m ³ 15 min TWA: 1 mg/m ³ 8 hr	
Starch		STEL: 30 mg/m ³ 15 min STEL: 12 mg/m ³ 15 min TWA: 10 mg/m ³ 8 hr TWA: 4 mg/m ³ 8 hr	

Exposure Controls

Engineering Measures

None under normal use conditions.

Personal protective equipment

Eye Protection

Wear safety glasses with side shields (or goggles)

Hand Protection

Protective gloves

Skin and body protection

Long sleeved clothing

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.

(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g.

sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

Respiratory Protection

No protective equipment is needed under normal use conditions

Recommended Filter type:

Particle filter

SAFETY DATA SHEET

Chocolate Agar

Revision Date 29-Mar-2023

Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice

Environmental exposure controls

No information available

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

Information on basic physical and chemical properties

Appearance

Physical State

Gel

Odor

No information available

Odor Threshold

No data available

pH

No information available

Melting Point/Range

No data available

Softening Point

No data available

Boiling Point/Range

No information available

Flash Point

No information available

Method - No information available

Evaporation Rate

No data available

Flammability (solid,gas)

No information available

Explosion Limits

No data available

Vapor Pressure

No data available

Vapor Density

No data available

(Air = 1.0)

Specific Gravity / Density

No data available

Bulk Density

No data available

Water Solubility

No information available

Solubility in other solvents

No information available

Partition Coefficient (n-octanol/water)

Component

log Pow

Octamethylcyclotetrasiloxane

6.488

Vitamin B12

3.57

L-Glutamine

-4.609

Adenine (6-Aminopurine)

-0.1

Thiazolium,

<-3.04

3-[(4-amino-2-methyl-5-pyrimidinyl)met

hyl]-5-(2-hydroxyethyl)-4-methyl-

chloride, monohydrochloride

Adenosine 5'-(trihydrogen

<-4

diphosphate), P'.fwdarw.5'-ester with

3-(aminocarbonyl)-1-.beta.-D-ribofuran

osylpyridinium, inner salt

Autoignition Temperature

No data available

Decomposition Temperature

No data available

Viscosity

No data available

Explosive Properties

No information available

Oxidizing Properties

No information available

SAFETY DATA SHEET

Chocolate Agar

Revision Date 29-Mar-2023

SECTION 10: STABILITY AND REACTIVITY

Reactivity

None known, based on information available.

Chemical Stability

Stable under normal conditions.

Possibility of Hazardous Reactions

Hazardous Polymerization Hazardous Reactions

Hazardous polymerization does not occur.
None under normal processing.

Conditions to Avoid

Incompatible products. Excess heat. Avoid dust formation.

Incompatible Materials

None known.

Hazardous Decomposition Products

None under normal use conditions.

SECTION 11: TOXICOLOGICAL INFORMATION

Information on Toxicological Effects

Acute Toxicity

Toxicology data for the components

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
Hydrogen chloride	900 mg/kg (Rabbit)	> 5010 mg/kg (Rabbit)	LC50 = 4701 ppm (rat) 30 min (gas), LC50 = 588 ppm (4h) by extrapolation LC50 = 8.3 mg/L (rat) 30 min (aerosols) (MMAD < 5µm)
Octamethylcyclotetrasiloxane	LD50 > 4800 mg/kg (Rat male)	LD50 > 2375 mg/kg (Rat)	LC50 = 36 mg/L (Rat) 4 h
Glucose	25.8 g/kg (Rat)		
L-Glutamine	LD50 = 7500 mg/kg (Rat)		
Adenine (6-Aminopurine)	LD50 = 227 mg/kg (Rat)		
p-Aminobenzoic acid	>6 g/kg (Rat)		
Iron(III) nitrate nonahydrate	LD50 = 3250 mg/kg (Rat)		
Thiazolium, 3-[(4-amino-2-methyl-5-pyrimidinyl) methyl]-5-(2-hydroxyethyl)-4-methyl-	LD50 = 3710 mg/kg (Rat)		

SAFETY DATA SHEET

Chocolate Agar

Revision Date 29-Mar-2023

chloride, monohydrochloride			
Sodium chloride	LD50 = 3 g/kg (Rat)	LD50 > 10000 mg/kg (Rabbit)	LC50 > 42 mg/L (Rat) 1 h
Dipotassium phosphate	8 g/kg (rat)	LD50 > 5000 mg/kg (Rabbit)	
Propanoic acid, 2-oxo-, sodium salt	5600 mg/kg (Rat)		
Agar	LD50 = 11 g/kg (Rat)		
Water	-	-	-

Chronic Toxicity

Carcinogenicity

There are no known carcinogenic chemicals in this product

Legend:

X - Listed '-' - Not Listed T - Indicates a substance that is the subject of a Section 4 test rule under TSCA. XU - Indicates a substance exempt from reporting under the Inventory Update Rule, i.e. Partial Updating of the TSCA Inventory Data Base Production and Site Reports (40 CFR 710(B))

Sensitization

No information available

Mutagenic Effects

No information available

Reproductive Effects

No information available

Developmental Effects

No information available

Target Organs

No information available.

Symptoms

Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation.

SECTION 12: ECOLOGICAL INFORMATION

Ecotoxicity effects

Contains no substances known to be hazardous to the environment or that are not degradable in waste water treatment plants.

Component	Freshwater Fish	Water Flea	Freshwater Algae	Microtox
Octamethylcyclotetrasiloxane	LC50: > 1000 mg/L, 96h (Lepomis macrochirus) LC50: > 500 mg/L, 96h (Brachydanio rerio)			
p-Aminobenzoic acid				= 27.4 mg/L EC50 Photobacterium phosphoreum 30 min 15 °C
Thiazolium, 3-[(4-amino-2-methyl-5-pyrimidinyl)methyl]- 5-(2-hydroxyethyl)-4-methyl- chloride, monohydrochloride	LC50 >100 mg/L/96h	EC50 >100 mg/L/48h		
Sodium chloride	Pimephals prome: LC50: 7650 mg/L/96h	EC50: 1000 mg/L/48h		

Persistence and degradability

No information available

Bioaccumulative potential

No information available

Component	log Pow	Bioconcentration factor (BCF)
Octamethylcyclotetrasiloxane	6.488	12400 dimensionless

SAFETY DATA SHEET

Chocolate Agar

Revision Date 29-Mar-2023

Vitamin B12	3.57	No data available
L-Glutamine	-4.609	No data available
Adenine (6-Aminopurine)	-0.1	No data available
Thiazolium, 3-[(4-amino-2-methyl-5-pyrimidinyl)methyl]- 5-(2-hydroxyethyl)-4-methyl- chloride, monohydrochloride	<-3.04	No data available
Adenosine 5'-(trihydrogen diphosphate), P'.fwdarw.5'-ester with 3-(aminocarbonyl)-1-.beta.-D-ribofuranosyl pyridinium, inner salt	<-4	No data available

Mobility in soil No information available. .

Other adverse effects No information available

SECTION 13: DISPOSAL CONSIDERATIONS

Waste treatment methods

Waste from Residues/Unused Products Dispose of in accordance with local regulations

Contaminated Packaging Empty containers should be taken to an approved waste handling site for recycling or disposal

SECTION 14: TRANSPORT INFORMATION

IMDG/IMO Not regulated

Road and Rail Transport Not regulated

IATA Not regulated

Special Precautions for User No special precautions required

SECTION 15: REGULATORY INFORMATION

Safety, health and environmental regulations/legislation specific for the substance or mixture

International Inventories X = listed

Component	EINECS	TSCA	DSL	PICCS	ENCS	ISHL	IECSC	AICS	KECL
Hydrogen chloride	-	X	X	X	X	X	X	X	KE-20189
Octamethylcyclotetrasiloxane	-	X	X	X	X	X	X	X	KE-26606
Hemoglobins	-	X	-	-	-	-	-	X	-
Glucose	-	X	X	X	X	X	X	X	KE-17727
Vitamin B12	-	X	X	X	-	X	X	X	KE-11218
L-Glutamine	-	X	X	X	X	X	X	X	KE-17780
Adenine (6-Aminopurine)	-	X	X	X	X	X	X	X	KE-29916
6H-Purin-6-one, 2-amino-1,7-dihydro-, monohydrochloride	-	X	-	-	-	-	X	X	-

SAFETY DATA SHEET

Chocolate Agar

Revision Date 29-Mar-2023

p-Aminobenzoic acid	205-753-0	X	X	X	X	X	X	X	KE-01199
Cysteine hydrochloride, L-(+)-, monohydrate	-	-	-	X	X		X	X	KE-01430
Thiazolium, 3-[(4-amino-2-methyl-5-pyrimidinyl)methyl]-4-methyl-5-(4,6,6-trihydroxy-3,5-dioxo-4,6-diphosphahex-1-yl)-, chloride, P,P'-dioxide	205-836-1	X	-	-	-		X	X	KE-01484
Iron(III) nitrate nonahydrate	-	-	-	X	X		X	X	-
Thiazolium, 3-[(4-amino-2-methyl-5-pyrimidinyl)methyl]-5-(2-hydroxyethyl)-4-methyl-, chloride, monohydrochloride	-	X	X	X	X	X	X	X	KE-01482
Adenosine 5'-(trihydrogen diphosphate), P',P''-ester with 3-(aminocarbonyl)-1-beta.-D-ribofuranosylpyridinium, inner salt	-	X	X	-	-		X	X	KE-25879
Peptones	-	X	X	X	X	X	X	X	KE-28131
Sodium chloride	-	X	X	X	X	X	X	X	KE-31387
Dipotassium phosphate	-	X	X	X	X	X	X	X	KE-12167
Starch	-	X	X	X	X	X	X	X	KE-32128
Propanoic acid, 2-oxo-, sodium salt	204-024-4	X	X	X	X	X	X	X	KE-27653
Yeast, ext.	-	X	X	X	-		X	X	KE-05-1355
Caseins, hydrolyzates	-	X	X	X	X	X	X	X	KE-05-0318
Gelatins, hydrolyzates	-	X	X	X	-		X	X	KE-17576
Agar	-	X	X	X	-		X	X	KE-00275
Water	231-791-2	X	X	X	X		X	X	KE-35400

Component	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements	Rotterdam Convention (PIC)	Basel Convention (Hazardous Waste)
Hydrogen chloride	25 tonne	250 tonne		Annex I - Y34

National Regulations

Persistent Organic Pollutant
Ozone Depletion Potential

This product does not contain any known or suspected substance
This product does not contain any known or suspected substance

SECTION 16: OTHER INFORMATION

Legend

CAS - Chemical Abstracts Service

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit

ACGIH - American Conference of Governmental Industrial Hygienists

RPE - Respiratory Protective Equipment

LC50 - Lethal Concentration 50%

POW - Partition coefficient Octanol:Water

TWA - Time Weighted Average

IARC - International Agency for Research on Cancer

LD50 - Lethal Dose 50%

EC50 - Effective Concentration 50%

SAFETY DATA SHEET

Chocolate Agar

Revision Date 29-Mar-2023

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road

IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code

OECD - Organisation for Economic Co-operation and Development

BCF - Bioconcentration factor

ICAO/IATA - International Civil Aviation Organization/International Air Transport Association

MARPOL - International Convention for the Prevention of Pollution from Ships

ATE - Acute Toxicity Estimate

VOC - (Volatile Organic Compound)

Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

Revision Date

29-Mar-2023

Revision Summary

Not applicable.

In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

End of Safety Data Sheet